Repeated-variable reduced substitutions give proper self-embeddings of free Steiner quasigroups

Silvia Barbina, Enrique Casanovas

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Editorial note. The already public normal-form PropositionMay19_1 is included as solution support and is not counted again. The new target concerns which substitutions induce proper self-embeddings, a different conclusion.

Editorial note. The final single-occurrence exclusion uses the explicit inverse-term LemmaJune9_2: if such a term were in the image generated by a,c, its inverse term would put b in that same image. This is the immediate closing implication of the author chain.

Editorial scope. One original proposition with its full author local proof chain; no new claim assigned to any support lemma.

Allowed standard prerequisites (editorial).

- Elementary universal algebra and well-founded recursion. Universal mapping property, generated subalgebras, uniqueness of a homomorphism on generators, and recursion/induction on well-orders.
- Finite combinatorial accounting on terms and triples. Term length and rank, counting blocks by their greatest vertex, and monotone nonnegative integer sequences.

A **Steiner triple system** (STS) of order λ is a pair $(M, \mathcal{B})$ where M is a set with λ elements, and $\mathcal{B}$ is a collection of 3-element subsets of M (the **blocks**) such that any two $x, y \in M$ are contained in exactly one block. If we require that any two $x, y \in M$ are contained in *at most* one block, M is said to be a **partial Steiner triple system**.

From a model theoretic standpoint, Steiner triple systems can be viewed as relational structures with a single a ternary relation R interpreted as the set of blocks (that is, three elements x, y and z are in a block if and only if $R(x, y, z)$). Then R must satisfy

- for all permutations $\sigma \in S_3$, $\forall x_1 x_2 x_3 \ \big(R(x_1, x_2, x_3) \to R(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)})\big)$
- $\forall xyz \, (R(x, y, z) \to y \neq z)$
- $\forall xywz \, (R(x, y, w) \wedge R(x, y, z) \to w = z)$.

With this choice of language, a substructure of a Steiner triple system is a partial Steiner triple system.

An STS $(M, \mathcal{B})$ determines a *Steiner quasigroup*, that is, a pair $(M, \cdot)$ where $\cdot$ is a binary operation such that

- $\forall x\,(x \cdot x = x)$
- if $x \neq y$, then $x \cdot y = z$ where $\{x, y, z\} \in \mathcal{B}$.

The operation $\cdot$ defined in this way satisfies

1. $\forall x\,(x \cdot y = y \cdot x)$
2. $\forall x\,(x \cdot x = x)$
3. $\forall x\,(x \cdot (x \cdot y) = y)$.

Conversely, any structure $(M, \cdot)$ where $\cdot$ satisfies properties 1, 2 and 3 determines an STS $(M, \mathcal{B})$.

In universal algebraic parlance, Steiner quasigroups are algebras. Thus, if M is a Steiner quasigroup, there are standard notions of a *subalgebra* of M, and of the subalgebra *generated* by a subset $A \subseteq M$. A subalgebra of a Steiner quasigroup is a substructure in the model theoretic sense, and in particular it is a Steiner quasigroup. If $A \subseteq M$, then $\langle A \rangle_M$ denotes the substructure of M generated by A. The subscript M is usually omitted when clear from the context.

A Steiner quasigroup induces a partial STS structure on each of its subsets, so we often refer to a partial Steiner triple system when we need to consider a subset of a Steiner quasigroup that is not a substructure.

Given a class $\mathcal{V}$ of algebras in a given signature, we say that a member U of $\mathcal{V}$ generated by a subset $X \subseteq U$ has the **universal mapping property** (UMP) over X with respect to $\mathcal{V}$ if for every $V \in \mathcal{V}$, every map

$$\varphi : X \to V$$

extends to a homomorphism $\hat{\varphi} : U \to V$. A standard argument proves that the homomorphism $\hat{\varphi}$ is unique. Such an algebra U is said to be **freely generated by** X, and X is a **set of free generators** of U.

It is well known that when $\mathcal{V}$ is a variety, given any cardinal $\lambda \geq 1$ there are $U \in \mathcal{V}$ and $X \subseteq U$ such that U is freely generated by X and $|X| = \lambda$.

Since the class $\mathcal{K}$ of Steiner quasigroups is a variety in the universal algebraic sense, for each cardinal $\lambda \geq 1$, there is a Steiner quasigroup that is freely generated by a set of cardinality λ. Then a **free Steiner quasigroup** is a Steiner quasigroup that is freely generated by some subset X, and so it has UMP over X with respect to the class $\mathcal{K}$. We assume $\lambda \geq 3$ to exclude the trivial cases of Steiner quasigroups with one and three elements.

Definition 0.1 Let $(M, \cdot)$ be a Steiner quasigroup. A subset $A \subseteq M$ is **independent** if the generated substructure $\langle A \rangle_M$ is freely generated from A, that is, if $\langle A \rangle_M$ has UMP over A for the class of Steiner quasigroups.

We say that $A \subseteq M$ is a **free base** from M if it is independent and it generates M.

When we say $\bar{a} = \langle a_i : i \in I \rangle$ is independent, we understand $a_i \neq a_j$ if $i, j \in I$ and $i \neq j$.

If M is a Steiner quasigroup generated by the subset A and $\bar{a}$ is an enumeration of A, then for every $b \in M$ there is a term $t(\bar{x})$ such that $b = t(\bar{a})$.

Let M be a Steiner quasigroup generated by $A \subseteq M$. Then $M = \bigcup_{n<\omega} S_n$ for a chain of sets S_n such that $S_0 = A$ and $S_{n+1} = \{a \cdot b \mid a, b \in S_n\}$. Note that $S_n \subseteq S_{n+1}$ since $a \cdot a = a$. We will write $S_{<n}$ for $\bigcup_{i<n} S_i$. The sets S_n are called the **levels** of M over A. An element $a \in M$ has **level** n **over** A if $a \in S_n \setminus S_{<n}$.

Definition 0.2 Let M be a Steiner quasigroup, and let $A \subseteq M$ be a free base. The sequence $(S_n \mid n < \omega)$ of levels described in Proposition 0.6 will be called a **standard free construction of** M **over** A.

Definition 0.3 We define an equivalence relation $\sim$ on the set of terms in such a way that $t_1 \sim t_2$ if t_2 is obtained from t_1 by one or more applications of commutativity to subterms.

A term is said to be **reduced** if it contains no occurrences of subterms of any of the following forms

- $t_1 t_2$ with $t_1 \sim t_2$
- $t_1(t_2 t_3)$ with $t_1 \sim t_2$ or $t_1 \sim t_3$
- $(t_1 t_2)t_3$ with $t_1 \sim t_3$ or $t_2 \sim t_3$.

Definition 0.4 The **rank** $\mathrm{rk}(t)$ of a term t is defined to be

- 0 if t is a variable
- $\max\{\mathrm{rk}(r), \mathrm{rk}(s)\} + 1$ if $t = r \cdot s$.

Proposition 0.5 Let $\bar{a}, b$ be independent, let $r(\bar{x}, y)$ be a reduced term with at least two occurrences of y, let $c = r(\bar{a}, b)$ and let $f : \langle \bar{a}, b\rangle \to \langle \bar{a}, c\rangle$ be the surjective homomorphism induced by $f(\bar{a}) = \bar{a}$ and $f(b) = c$. Then f is an isomorphism between $\langle \bar{a}, b\rangle$ and $\langle \bar{a}, c\rangle$ and $b \notin \langle \bar{a}, c\rangle$. In fact, for every term $t(\bar{x}, y)$ having exactly one occurrence of y, $t(\bar{a}, b) \notin \langle \bar{a}, c\rangle$.

Proposition 0.6 Let M be a Steiner quasigroup and let $A \subseteq M$. The following are equivalent:

1. A is a free base of M.
2. M is generated by A with levels $(S_n \mid n < \omega)$ such that $A = S_0$ and
   (a) if $a, b \in S_n \setminus S_{<n}$ and $a \neq b$, then $a \cdot b \notin S_n$.
   (b) if $a \in S_{n+1} \setminus S_n$, there is a unique pair $\{b, c\} \subseteq S_n$ such that $b \neq c$ and $a = b \cdot c$.

Moreover, property (b) implies that if $a \in S_{n+1} \setminus S_n$, and $b, a \cdot b \in S_n$, then $\{b, a \cdot b\}$ is the unique pair given by (b) for a.

Proof. $2 \Rightarrow 1$. Let N be a Steiner quasigroup and let $f : A \to N$ be a mapping. We extend f to a homomorphism $\hat{f} : M \to N$. We obtain $\hat{f}$ as the union $\hat{f} = \bigcup_{n<\omega} f_n$ of a chain of partial homomorphisms $f_n : S_n \to N$. This means that $f_n(a \cdot b) = f_n(a) \cdot f_n(b)$ whenever a, b and $a \cdot b$ belong to $\mathrm{dom}(f_n) = S_n$. We start with $f_0 = f$. Recall that $A = S_0$ and that $a \cdot b \notin S_0$ if $a, b \in S_0$ and, therefore, f_0 is a partial homomorphism with domain S_0. Now we extend

$f_n : S_n \to N$ to f_{n+1}. For $a \in S_{n+1} \setminus S_n$, there is a unique pair $\{b, c\} \subseteq S_n$ of distinct elements b, c with $a = b \cdot c$. We let $f_{n+1}(a) = f_n(b) \cdot f_n(c)$. By (a) and (b), the only defined products involving a in S_{n+1} are $a = b \cdot c = c \cdot b$, $a \cdot a = a$, $a \cdot b = b \cdot a = c$ and $a \cdot c = c \cdot a = b$ and f_{n+1} respects these products.

$1 \Rightarrow 2$. It is easy to construct a Steiner quasigroup $M' \supseteq A$ with the properties described in 2. It is enough to start with $S'_0 = A$ and inductively define S'_{n+1} by adding to S'_n a different element $a \cdot b$ for each pair $\{a, b\}$ of elements of S'_n whose product is not defined in S'_n. The only products involving the new element $c = a \cdot b$ in S'_{n+1} are $c = a \cdot b = b \cdot a$, $c \cdot c = c$, $c \cdot a = a \cdot c = b$ and $c \cdot b = b \cdot c = a$. We have proven that A is a free base of $M' = \bigcup_{n<\omega} S'_n$. The uniqueness of free Steiner quasigroups over a given set of generators guarantees that M and M' are isomorphic over A. The isomorphism can be used to obtain the levels S_n in M with the required properties. □

Lemma 0.7

1. If $t_1 \cdot t_2 \sim r_1 \cdot r_2$, then either $t_1 \sim r_1$ and $t_2 \sim r_2$, or $t_1 \sim r_2$ and $t_2 \sim r_1$.
2. if $t \sim r$ and t' is a subterm of t then $t' \sim r'$ for some subterm r' of r.
3. If $t_1 \sim t_2$, then $\mathrm{rk}(t_1) = \mathrm{rk}(t_2)$ and $t_1(\bar{a}) = t_2(\bar{a})$ for any tuple $\bar{a}$.
4. If $t_1 \sim t_2$ and t_1 is reduced, then t_2 is reduced.
5. Any subterm of a reduced term is reduced.
6. Assume t, t' are reduced, $t \not\sim t'$ and there is no term r such that $t \sim t' \cdot r$ or $t' \sim t \cdot r$. Then $t \cdot t'$ is reduced.

Proof. 1 can be easily checked by induction on the number of applications of commutativity needed for $t_1 \cdot t_2 \sim r_1 \cdot r_2$. 2 is proved by induction on t. 3 is clear. 4 follows from 2. 5 and 6 are clear. □

The next result clarifies how the levels in a construction of a Steiner quasigroup M are linked to the ranks of the reduced terms giving the elements of M.

Lemma 0.8 Let $M = \langle \bar{a} \rangle$ be a Steiner quasigroup, with a construction with levels $(S_n \mid n < \omega)$.

1. If t is a term such that $\mathrm{rk}(t) \leq n$, then $t(\bar{a}) \in S_n$.
2. If $b \in S_n$, then $b = t(\bar{a})$ for some reduced term t of rank $\mathrm{rk}(t) \leq n$.
3. $S_n = \{t(\bar{a}) \mid \mathrm{rk}(t) \leq n\} = \{t(\bar{a}) \mid t \text{ is reduced and } \mathrm{rk}(t) \leq n\}$.

Proof. 1 is proved by induction on n. 3 follows from 1 and 2. We prove 2 by induction on n. Assume $n = 0$. If $\bar{a} = \langle a_i : i \in I \rangle$, then $b = a_k$ for some $k \in I$ and $t = x_k$ (a variable, hence reduced and of rank 0) is the required term. For the inductive step, assume $b \in S_{n+1}$. If $b \in S_n$ we apply the induction hypothesis. If $b \notin S_n$, then $b = b_1 \cdot b_2$ for $b_1, b_2 \in S_n$. By induction hypothesis, there are reduced terms t_1, t_2 of rank $\leq n$ such that $b_1 = t_1(\bar{a})$ and $b_2 = t_2(\bar{a})$. If $t_1 \sim t_2$, then $t_1(\bar{a}) = t_2(\bar{a})$ and then $b = b_1 \cdot b_2 = b_1 \cdot b_1 = b_1 \in S_n$, a contradiction. Hence, $t_1 \not\sim t_2$. If there is a term r such that $t_1 \sim t_2 \cdot r$, then $b = b_1 \cdot b_2 = (t_2(\bar{a}) \cdot r(\bar{a})) \cdot b_2 = (b_2 \cdot r(\bar{a})) \cdot b_2 = r(\bar{a})$. But $\mathrm{rk}(r) < \mathrm{rk}(t_2 \cdot r) =$

$\mathrm{rk}(t_1) \leq n$, which implies $b \in S_n$, a contradiction. A similar contradiction follows from the assumption that $t_2 \sim t_1 \cdot r$ for some term r. By Lemma 0.7, $t = t_1 \cdot t_2$ is reduced. Moreover $\mathrm{rk}(t) \leq n+1$ and $b = t(\bar{a})$. □

Proposition 0.9 Let $M = \langle \bar{a} \rangle$ be a free Steiner quasigroup with standard free construction $(S_i \mid i < \omega)$, let $\bar{a} = \langle a_i : i \in I \rangle$ be independent, and let $t = t(\bar{x})$ be a reduced term of rank k. Then

1. $t(\bar{a}) \in S_k \setminus S_{<k}$
2. if r is reduced and $r(\bar{a}) = t(\bar{a})$, then $r \sim t$.

Proof. We prove 1 and 2 jointly by induction on k, but with the extra assumption that $\mathrm{rk}(r) = \mathrm{rk}(t)$. This assumption can be eliminated later since 1 excludes other possibilities.

Base case $k = 0$. If $\mathrm{rk}(t) = 0$, then t is a variable x_i and $t(\bar{a}) = a_i \in S_0$. Moreover, if r has rank 0, r is a variable x_j and $t(\bar{a}) = r(\bar{a})$ implies $a_i = a_j$ and therefore $x_i = x_j$. Hence $t \sim r$.

Inductive step $k+1$. Let $t = t_1 \cdot t_2$, with $k_1 = \mathrm{rk}(t_1) \leq \mathrm{rk}(t_2) = k$ and let $b = t(\bar{a})$. By inductive hypothesis, $b_1 = t_1(\bar{a}) \in S_{k_1} \setminus S_{<k_1}$ and $b_2 = t_2(\bar{a}) \in S_k \setminus S_{<k}$. Then $b = b_1 \cdot b_2$ and there are different cases.

Case 1. Assume $k_1 = k$. If $b_1 = b_2$, then $b = b_1 = b_2$ and by inductive hypothesis $t_1 \sim t_2$. But then t is not reduced. Hence, $b_1 \neq b_2$. By construction, $b \in S_{k+1} \setminus S_k$. Moreover, if r is reduced of rank $k+1$ with $r(\bar{a}) = b$, then $r = r_1 \cdot r_2$, with r_1, r_2 reduced and, without loss of generality, $\mathrm{rk}(r_1) \leq \mathrm{rk}(r_2) = k$. By inductive hypothesis $b_1' = r_1(\bar{a}) \in S_{\mathrm{rk}(r_1)} \setminus S_{<\mathrm{rk}(r_1)}$ and $b_2' = r_2(\bar{a}) \in S_k \setminus S_{<k}$. Then $b = b_1' \cdot b_2'$ and by construction $\{b_1, b_2\} = \{b_1', b_2'\}$. If $b_1 = b_1'$ and $b_2 = b_2'$, then $\mathrm{rk}(r_1) = k_1$ and by inductive hypothesis $t_1 \sim r_1$ and $t_2 \sim r_2$. Therefore in this case $t \sim r$. If $b_1 = b_2'$ and $b_2 = b_1'$, then $k = k_1 = \mathrm{rk}(r_1)$ and, by inductive hypothesis, $t_1 \sim r_2$ and $t_2 \sim r_1$. Again we conclude that $t \sim r$.

Case 2. Assume $k_1 < k$ and $b \notin S_{<k}$. Note that $b \neq b_2$ and $b \notin S_k$. By construction, $b = b_1 \cdot b_2 \in S_{k+1}$. Now let r be reduced, of rank $k+1$ and such that $r(\bar{a}) = t(\bar{a}) = b$. Exactly as in the previous case we conclude that $r \sim t$.

Case 3. The last case is $k_1 < k$ and $b \in S_j \setminus S_{<j}$ for some $j < k$. We show this is impossible. We know that there is some reduced term s of rank j such that $s(\bar{a}) = b$. Now we claim that $s \cdot t_1$ is reduced. Otherwise, there are two possible cases:

(i) $s \sim t_1$. But then $b = b_1$ and hence $b = b_1 = b_2$, which is not the case.

(ii) $t_1 \sim s \cdot r$ or $s \sim t_1 \cdot r$ for some (reduced) term r. Then $r(\bar{a}) = t_1(\bar{a}) \cdot s(\bar{a}) = b_1 \cdot b = b_2$ and by inductive hypothesis $\mathrm{rk}(r) = k$, contradicting $\mathrm{rk}(t_1) = k_1 < k$ and $\mathrm{rk}(s) = j < k$.

Therefore, we conclude that $s \cdot t_1$ is reduced. Then, since $\mathrm{rk}(s \cdot t_1) \leq k$ and $s \cdot t_1(\bar{a}) = b \cdot b_1 = b_2$, we can apply the inductive hypothesis to get $\mathrm{rk}(s \cdot t_1) = k$ and $s \cdot t_1 \sim t_2$. Then $t \sim t_1 \cdot t_2 \sim t_1 \cdot (s \cdot t_1)$, which says that t is not reduced. □

Lemma 0.10 If $t = t(\bar{x}, y)$ has only one occurrence of y, then there is a term $r(\bar{x}, z)$ such that the equations $t(\bar{x}, y) = z$ and $r(\bar{x}, z) = y$ are equivalent in every Steiner quasigroup.

Proof. Induction on t. If t is a variable, $t = y$ and we can take $r = z$. If $t = t_1 \cdot t_2$, without loss of generality, y appears in t_1 and not in t_2. Then $t(\bar{x}, y) = z$ is equivalent to $t_1(\bar{x}, y) = t_2(\bar{x}) \cdot z$. By induction hypothesis, there is a term $r(\bar{x}, u)$ for which $t_1(\bar{x}, y) = u$ is equivalent to $r(\bar{x}, u) = y$. Then $t_1(\bar{x}, y) = t_2(\bar{x}) \cdot z$ is equivalent to $r(\bar{x}, t_2(\bar{x}) \cdot z) = y$. □

Proposition 0.11 If $t = t(\bar{x}, y)$ has only one occurrence of y and $\bar{a}b$ is independent, where $\bar{a} = a_1, \ldots, a_n$, then the mapping induced by

- $a_i \mapsto a_i$ for $i = 1, \ldots, n$
- $b \mapsto t(\bar{a}, b)$

is an automorphism of $\langle \bar{a}b \rangle$.

Proof. Since $\bar{a}b$ is independent, there is a homomorphism $f : \langle \bar{a}b \rangle \to \langle \bar{a}b \rangle$ such that $f(\bar{a}) = \bar{a}$ and $f(b) = t(\bar{a}, b)$. We claim that there is another homomorphism $g : \langle \bar{a}b \rangle \to \langle \bar{a}b \rangle$ such that $g(\bar{a}) = \bar{a}$ and $g(t(\bar{a}, b)) = b$. Then $g \circ f : \langle \bar{a}b \rangle \to \langle \bar{a}b \rangle$ is a homomorphism such that $g \circ f(\bar{a}) = \bar{a}$ and $g \circ f(b) = b$. Hence $g \circ f$ is the identity on $\langle \bar{a}b \rangle$, which implies that f is one-to-one. Lemma 0.10 shows that $b \in \langle \bar{a}, t(\bar{a}, b) \rangle$, which is the range of f. Hence f is surjective.

We prove the claim by induction on t. If t is a single variable, it must be $t = y$ and then $f(b) = b$ and we take $g = f$. Assume $t = t_1 \cdot t_2$. Without loss of generality, y occurs in t_1 and not in t_2. The induction hypothesis gives a homomorphism $h : \langle \bar{a}b \rangle \to \langle \bar{a}b \rangle$ such that $h(\bar{a}) = \bar{a}$ and $h(t_1(\bar{a}, b)) = b$. Let $a = t_2(\bar{a})$ and note that $h(a) = a$ and $h(t(\bar{a}, b)) = h(t_1(\bar{a}, b)) \cdot h(t_2(\bar{a})) = b \cdot a$. Let $j : \langle \bar{a}b \rangle \to \langle \bar{a}b \rangle$ be the homomorphism induced by $j(\bar{a}) = \bar{a}$ and $j(b) = b \cdot a$. Then $j(b \cdot a) = (b \cdot a) \cdot a = b$ and hence $j \circ h(t(\bar{a}, b)) = j(b \cdot a) = b$ and $j \circ f(\bar{a}) = \bar{a}$. Therefore $g = j \circ h$ satisfies the requirements for t. □

Lemma 0.12 Assume $\bar{x} = \langle x_i : i \in I \rangle$ is a tuple of variables and y, z, x_i are pairwise distinct. Let $t_i = t_i(\bar{x}, z)$ for $i = 1, 2$, let $r = r(\bar{x}, y)$ and assume y occurs in r. We write $t_i' = t_i(\bar{x}, r(\bar{x}, y))$ for the substitution, and similarly for other terms. If $t_1' \sim t_2'$, then $t_1 \sim t_2$.

Proof. By induction on t_1.

(i) Let $t_1 = x_i$. Then $t_1' = x_i \sim t_2'$, which implies $t_2' = x_i$. Since y occurs in r, $t_2 = x_i$ and hence $t_1 \sim t_2$.

(ii) Let $t_1 = z$. Then $t_1' = r \sim t_2'$, which implies that y occurs in t_2' and, therefore, that z occurs in t_2. If $t_2 \neq z$, then $\text{length}(t_2') > \text{length}(r) = \text{length}(t_1')$, which is impossible if $t_1' \sim t_2'$. Thus, $t_2 = z$ and then $t_1 \sim t_2$.

(iii) Let $t_1 = s_1 \cdot s_2$ for some terms, s_1, s_2. Then $t_1' = s_1' \cdot s_2' \sim t_2'$. Note that $t_2 \neq x_i$, since otherwise $t_2' = t_2 = x_i \not\sim s_1' \cdot s_2'$. In case $t_2 = z$, we get $t_2' = r$, but since z is in t_1 (because y occurs in t_1'), then $\text{length}(t_1') > \text{length}(r) = \text{length}(t_2')$, contradicting $t_1' \sim t_2'$. Hence, $t_2 \neq z$. Therefore, $t_2 = u_1 \cdot u_2$ for some terms u_1, u_2. Therefore, $t_2' = u_1' \cdot u_2' \sim s_1' \cdot s_2'$. Without loss of generality, $u_1' \sim s_1'$ and $u_2' \sim s_2'$. By inductive hypothesis, $u_1 \sim s_1$ and $u_2 \sim s_2$. Then $t_1 = s_1 \cdot s_2 \sim u_1 \cdot u_2 = t_2$. □

Lemma 0.13 Assume $\bar{x} = \langle x_i : i \in I \rangle$ is a tuple of variables and y, z, x_i are pairwise distinct. Let $t = t(\bar{x}, z)$, let $r = r(\bar{x}, y)$ and assume y occurs in r. As above, we write $t' = t(\bar{x}, r(\bar{x}, y))$ for the substitution. Assume $r = r_1 \cdot r_2$ and y occurs in each r_i. If t and r are reduced, then t' is reduced.

Proof. By induction on t.

(i) Let $t = x_i$. Then $t' = t$ is reduced.

(ii) Let $t = z$. Then $t' = r$ is reduced.

(iii) Let $t = t_1 \cdot t_2$ for some terms t_1, t_2. By inductive hypothesis, each $t'_i = t_i(\bar{x}, r(\bar{x}, y))$ is reduced. If $t' = t'_1 \cdot t'_2$ is not reduced, then either $t'_1 \sim t'_2$ or $t'_1 \sim t'_2 \cdot s$ for some term s or $t'_2 \sim t'_1 \cdot s$ for some term s. If $t'_1 \sim t'_2$, we can apply Lemma 0.12 and obtain $t_1 \sim t_2$, which implies that t is not reduced. Now assume, without loss of generality, that $t'_1 \sim t'_2 \cdot s$ for some term s. We consider three subcases:

Subcase (a): $t_1 = x_i$. Then $t'_1 = x_i \sim t'_2 \cdot s$, which is impossible by length considerations.

Subcase (b): $t_1 = z$. Then $t'_1 = r_1 \cdot r_2 \sim t'_2 \cdot s$. The case $r_1 \sim t'_2$ is not possible, since it implies that z occurs in t_2 and then $\mathrm{length}(t'_2) \geq \mathrm{length}(r) > \mathrm{length}(r_1)$. Hence $r_1 \sim s$ and $r_2 \sim t'_2$. But again this is impossible, because it implies that z occurs in t_2 and $\mathrm{length}(t'_2) \geq \mathrm{length}(r) > \mathrm{length}(r_2)$.

Subcase (c): $t_1 = s_1 \cdot s_2$ for some terms s_1, s_2. Then $t'_1 = s'_1 \cdot s'_2 \sim t'_2 \cdot s$. Without loss of generality, $s'_1 \sim t'_2$ and $s'_2 \sim s$. By Lemma 0.12, $s_1 \sim t_2$. Then $t_1 \sim t_2 \cdot s_2$ and then $t \sim (t_2 \cdot s_2) \cdot t_2$ is not reduced. □

Proof. Consider first the case $r = r_1 \cdot r_2$, where each r_i contains an occurrence of y. In this case, by Lemma 0.13, for every reduced term $t(\bar{x}, z)$, the term $t'(\bar{x}, y) = t(\bar{x}, r(\bar{x}, y))$ is reduced.

We prove first that f is one-to-one. Assume $f(c_1) = f(c_2)$. For each i there is some reduced term $t_i(\bar{x}, z)$ such that $c_i = t_i(\bar{a}, b)$. Since the terms t'_1, t'_2 are reduced and $t'_1(\bar{a}, b) = f(c_1) = f(c_2) = t'_2(\bar{a}, b)$, by Proposition 0.9, $t'_1 \sim t'_2$. Then $t_1 \sim t_2$ by Lemma 0.12. Then $c_1 = t_1(\bar{a}, b) = t_2(\bar{a}, b) = c_2$.

Now we prove that $b \notin \langle \bar{a}, c \rangle$. Assume $b \in \langle \bar{a}, c \rangle$. Then there is a reduced term $t(\bar{x}, z)$ such that $b = t(\bar{a}, c)$. Note that z occurs in t, since otherwise $b \in \langle \bar{a} \rangle$. By Lemma 0.13, $t'(\bar{x}, y) = t(\bar{x}, r(\bar{x}, y))$ is reduced. Clearly, $b = t'(\bar{a}, b)$. If $k = \mathrm{rk}(t')$, Proposition 0.9 gives $t'(\bar{a}, b) \in S_k(\bar{a}, b) \setminus S_{<k}(\bar{a}, b)$. But $k \geq \mathrm{rk}(r) > 0$ and $b \in S_0(\bar{a}, b)$, a contradiction.

Now we consider the general case of a term $r(\bar{x}, y)$ with at least two occurrences of y. We claim that f can be written as a composition $f = g \circ h$, where

(i) h is an isomorphism between $\langle \bar{a}, b \rangle$ and $\langle \bar{a}, d \rangle$, where $d = s(\bar{a}, b)$ for a reduced term $s(\bar{x}, y)$ which is a product $s = s_1 \cdot s_2$ of terms s_i containing y, and h is induced by $h(\bar{a}) = \bar{a}$ and $h(b) = s(\bar{a}, b) = d$, and

(ii) g is an automorphism of $\langle \bar{a}, d \rangle = \langle \bar{a}, c \rangle$ and $c = t(\bar{a}, d)$ for a term $t(\bar{x}, y)$ with exactly one occurrence of y and g is induced by $g(\bar{a}) = \bar{a}$ and $g(d) = t(\bar{a}, d) = c$.

We prove this by induction on r. Clearly, r is not a variable, hence $r = r_1 \cdot r_2$ for some reduced terms $r_i = r_i(\bar{x}, y)$. If each r_i contains an occurrence of y, we are in the case we have discussed at the beginning and we can take $f = h$ and take g to be the identity on $\langle \bar{a}, c \rangle$. If one of the terms r_i, say r_1, does not contain y, then $r_1 = r_1(\bar{x})$ and $r_2(\bar{x}, y)$ contains at least two occurrences of y. By inductive hypothesis for r_2, we obtain an isomorphism $f^* : \langle \bar{a}, b \rangle \to \langle \bar{a}, c^* \rangle$, induced by $f^*(\bar{a}) = \bar{a}$ and $f^*(b) = c^* = r_2(\bar{a}, b)$ which can be decomposed as $f^* = g^* \circ h$, where $h : \langle \bar{a}, b \rangle \to \langle \bar{a}, d \rangle = \langle \bar{a}, c^* \rangle$ is induced by $h(\bar{a}) = \bar{a}$ and $h(b) = s(\bar{a}, b) = d$, where $s = s_1 \cdot s_2$ is reduced and each s_i contains an occurrence of y, and g^* is an automorphism of $\langle \bar{a}, d \rangle$ induced by $g^*(\bar{a}) = \bar{a}$ and

$g^*(d) = t^*(\bar{a}, d) = c^*$, where $t^*(\bar{x}, y)$ contains a unique occurrence of y. Now $t = r_1 \cdot t^*$ is a term with a unique occurrence of y and $t(\bar{a}, d) = r_1(\bar{a}) \cdot t^*(\bar{a}, d) = r_1(\bar{a}) \cdot c^* = r_1^*(\bar{a}) \cdot r_2(\bar{a}, b) = r(\bar{a}, b) = c$. Hence, there is an automorphism g of $\langle \bar{a}, d \rangle$ induced by $g(\bar{a}) = \bar{a}$ and $g(d) = t(\bar{a}, d) = c$ and $f = g \circ h$ is the desired decomposition. □

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